maddy1112-exe / STEPPER-MOTOR-INTERFACING

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STEPPER-MOTOR-INTERFACING / README.md

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Preview Code Blame 78 lines (53 loc) · 2.87 KB Raw

STEPPER MOTOR INTERFACING

AIM

To write an assembly language program in 8086 to rotate the motor at different speeds.

APPARATUS REQUIRED

S. No	Item	Specification	Quantity
1	Microprocessor kit	8086	1
2	Power Supply	+5 V DC, +12 V DC	1
3	Stepper Motor Interface board	-	1

S. No	Item	Specification	Quantity
4	Stepper Motor	-	1

THEORY

A motor in which the rotor is able to assume only discrete stationary angular positions is a **stepper motor**. The rotary motion occurs in a stepwise manner from one equilibrium position to the next.

Two-phase scheme: Any two adjacent stator windings are energized. There are two magnetic fields active in quadrature and none of the rotor pole faces can be in direct alignment with the stator poles. A partial but symmetric alignment of the rotor poles is of course possible.

ALGORITHM

For running the stepper motor in clockwise and anticlockwise directions:

1. Get the first data from the lookup table.
2. Initialize the counter and move data into the accumulator.
3. Drive the stepper motor circuitry and introduce delay.
4. Decrement the counter. If not zero, repeat from step (iii).
5. Repeat the above procedure both for backward and forward directions.

SWITCHING SEQUENCE OF STEPPER MOTOR

Memory Location	A1	A2	B1	B2	Hex Code
1200	1	0	0	0	09H
1201	0	1	0	1	05H
1202	0	1	1	0	06H
1203	1	0	1	0	0AH

PROGRAM

```
; Stepper Motor Interfacing Program in 8086 Assembly

START:   MOV DI, 1200H        ; Initialize memory location to store array
         MOV CX, 0004H        ; Initialize array size

DOWN1:   MOV AL, [DI]         ; Copy the first data into AL
         OUT C0, AL           ; Send it through port address

         MOV DX, 1010H        ; Delay subroutine
L1:      DEC DX
         JNZ L1

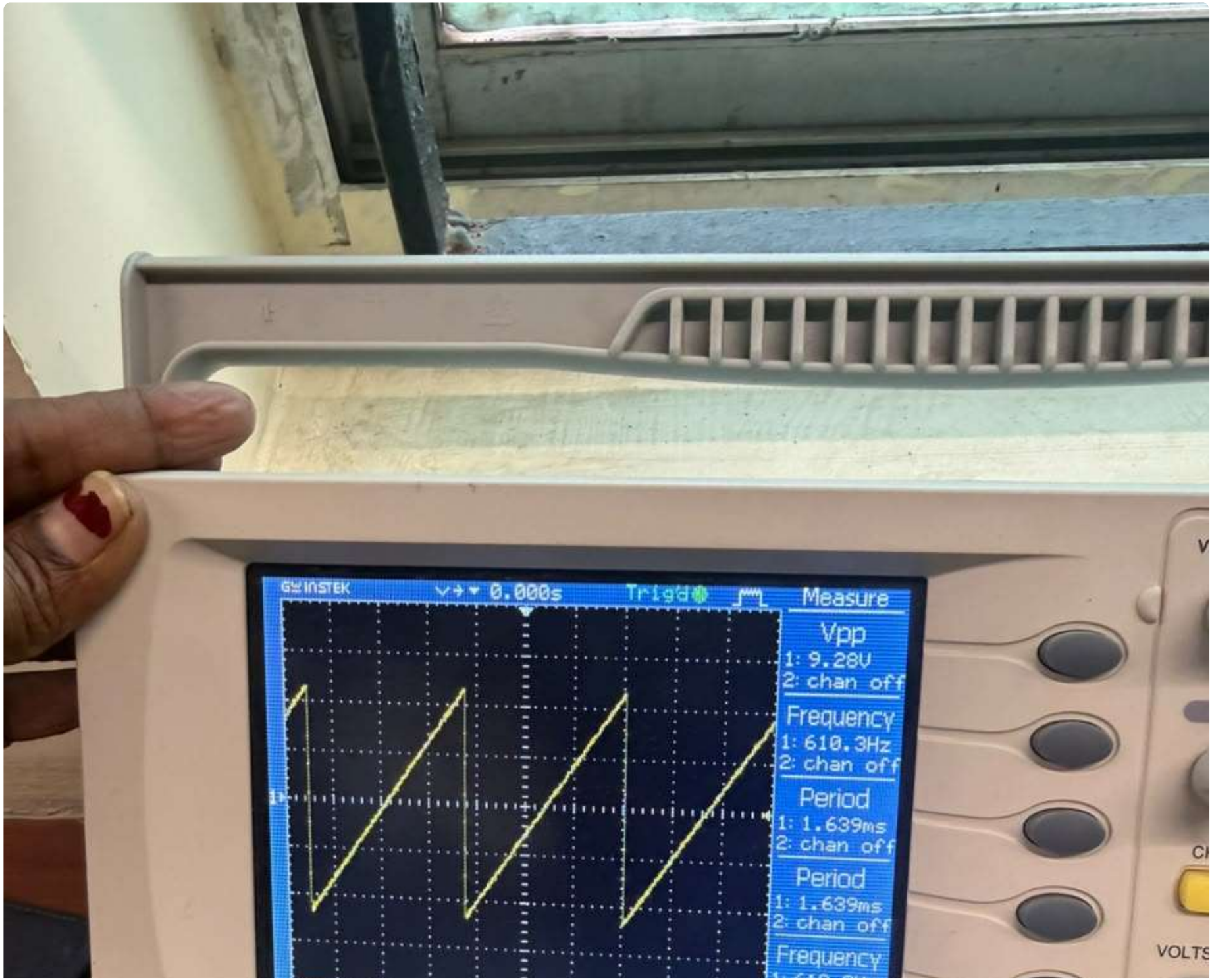
         INC DI               ; Go to next memory location
         LOOP DOWN1           ; Repeat until all data is sent

         JMP START            ; Continuous rotation

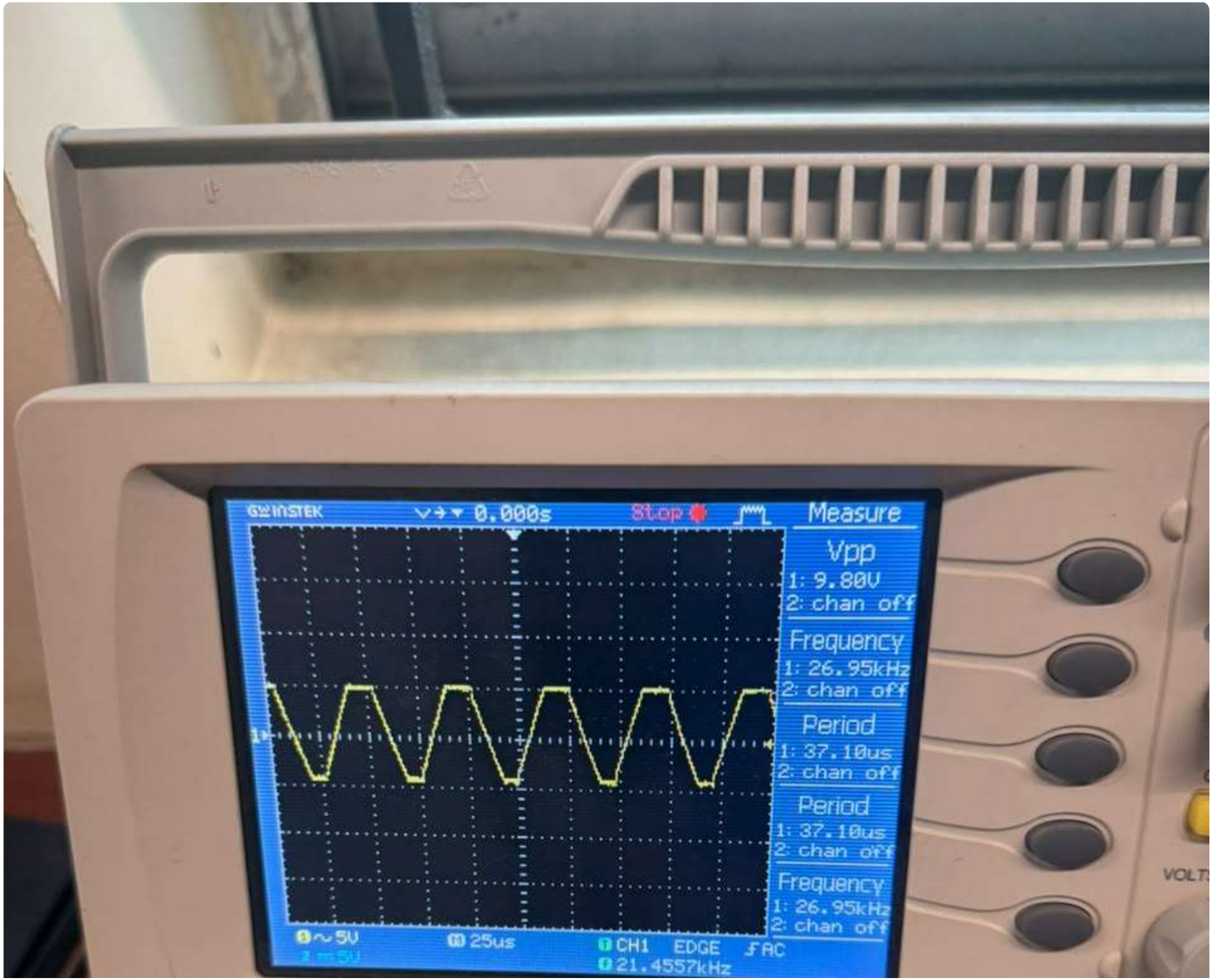
         HLT                  ; Stop
```

```
DATA:    DB 09H, 05H, 06H, 0AH ; Array of data
```

OUTPUT OF THE PROGRAM:


GW INSTEK
0.000s
Trig'd
Measure
Vpp
1: 9.28V
2: chan off
Frequency
1: 610.3Hz
2: chan off
Period
1: 1.639ms
2: chan off
Period
1: 1.639ms
2: chan off
Frequency
VOLTS


2V
500us
CH1 EDGE AC
610.358Hz
2: chan off
GW INSTEK GDS-1052-U
Digital Storage Oscilloscope
50MHz 250M Sa/s
CH1


GW INSTEK
0.000s
Stop
Measure
Vpp
1: 9.80V
2: chan off
Frequency
1: 26.95kHz
2: chan off
Period
1: 37.10us
2: chan off
Period
1: 37.10us
2: chan off
Frequency
1: 26.95kHz
2: chan off
5V
25us
CH1 EDGE AC
21.4557kHz



RESULT

Thus, the assembly language program for rotating the stepper motor in both clockwise and anticlockwise directions was written and verified.